

Data Challenge Spring 2025 Report

Dinsabqi Nor Azmi

Mukhriz Hazli

Eirfann Muzri

Aimin Tajuddin

1. Introduction

This challenge positions us as data consultants advising a new restaurant owner using a Yelp dataset of nearly 150,000 businesses and 7 million reviews. Our analysis targets two questions: (1) Which restaurant attributes are most strongly associated with business “success”, and which five should the owner invest in before opening? (2) Can we accurately predict whether a restaurant will remain open using operational, review, and user-engagement signals?

2. Data

We worked with five Yelp .parquet files: `business_y_removed.parquet`, `review.parquet`, `user.parquet`, `tip.parquet`, and `checkin.parquet`. Restaurants were filtered using the `categories` column and merged with review and tip information by users. Feature engineering included operational features (`days_open`, `avg_daily_hours`), attribute-based features (`top10_attr_count`, `attr_weighted_score`), review features (`avg_stars_review`, `total_reviews`, sentiment counts for positive, neutral, and negative reviews), and a location feature (`city_status` based on restaurant density). After merging and feature creation, we obtained 137,168 restaurants: 86,327 open, 23,397 closed, and 27,444 unlabeled held-out records used for prediction.

3. Method Analysis

3.1 Success Definition

We define business success as a composite of three conditions: `is_open = 1`, `stars  $\geq$  4.0`, and `review_count  $\geq$  20`. This removes large chains (which rarely close regardless of quality) and new restaurants with too few reviews to be statistically reliable.

3.2 Attribute Extraction & Selection

Restaurant attributes were extracted from nested dictionaries and flattened into 81 binary indicators. We retained the 21 attributes with prevalence $\geq 5\%$ among labeled restaurants. Each attribute was evaluated using percentage point (PP) differences in success rates and chi-square tests of independence. Attributes were ranked using a composite score combining PP-difference rank and chi-square rank.

Table 1: Top 5 Recommended Attributes

Rk	Attribute	PP Diff.	χ^2	Composite Score
1	Ambience_casual	+19.5pp	1650.3	1.5
2	BusinessParking_street	+17.9pp	2503.1	1.5
3	GoodForMeal_lunch	+17.5pp	743.0	3.0
4	OutdoorSeating	+11.6pp	675.3	4.0
5	BikeParking	+8.5pp	411.2	5.0

`Ambience_casual` and `BusinessParking_street` share the top composite rank (1.5), with the former showing the largest PP lift (+19.5pp) and the latter the strongest chi-square signal. All five attributes are statistically significant ($p < .001$).

4. Analysis: Predictive Modeling

4.1 Setup

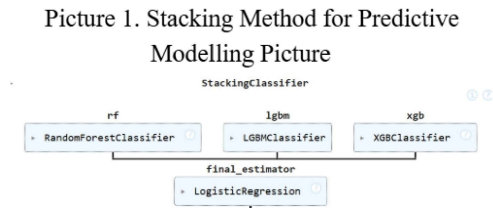
We predict the operational status (`is_open`) of 27,444 unlabeled restaurants using 13 engineered features including `stars`, `review_count`, `days_open`, `avg_daily_hours`, `avg_stars_review`, `attr_`

weighted_score, sentiment features, total_reviews, and city_status. The 109,724 labeled restaurants were split 80/20 stratified by is_open. Class imbalance (78.7% open) was handled using class_weight="balanced" and models were tuned using RandomizedSearchCV (5-fold CV, ROC-AUC scoring).

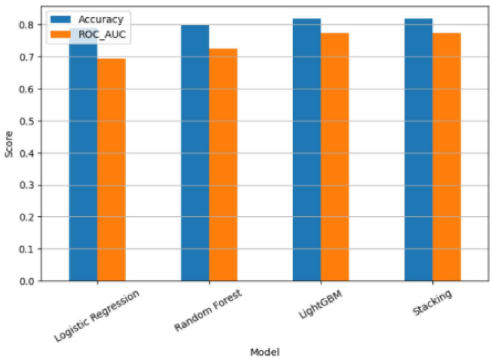
4.2 Model Results

Table 2: Model Comparison — 20% held-out test set (n = 21,945)

Model	Accuracy	ROC-AUC	Recall (Closed)
Stacking Ensemble (tuned)	81.8%	0.771	0.29



Picture 2. Model Performance Comparison



The stacking ensemble (base models: Random Forest, LightGBM, and XGBoost; meta-learner: Logistic Regression) was selected as the final model due to its strongest AUC (0.771) and accuracy (81.8%). Hyperparameter tuning improved class separation, increasing the recall for closed restaurants while slightly reducing overall accuracy, which is expected when dealing with imbalanced datasets. Applied to the unlabeled restaurants, the model predicted 24,911 open and 2,411 closed businesses.

5. Findings

Among the 21 attributes with more than 5% prevalence, our statistical analysis identified casual ambience (+19.5pp), street parking (+17.9pp), and lunch-friendly services (+17.5pp) as having the largest practical effects on business success. Outdoor seating and bike parking also provided moderate but statistically significant improvements.

Table 3: Top 3 States with the Highest Number of Restaurant Records

State	Best 5 Attributes (For All)	Best 3 Categories (For each State)
PA	Casual Ambience, Street Parking, Lunch-Friendly, Outdoor Seating, Bicycle Parking	Sandwiches, Pizza, Night Life
FL	—Same—	Night Life, American (Traditional), Sandwiches
TN	—Same—	Night Life, American (Traditional), Fast Food

References

[1] Yelp Inc. (2024). *Yelp Open Dataset*. <https://www.yelp.com/dataset>
[2] Rice, J. A. (2017). *Mathematical Statistics and Data Analysis*. Cengage Learning.